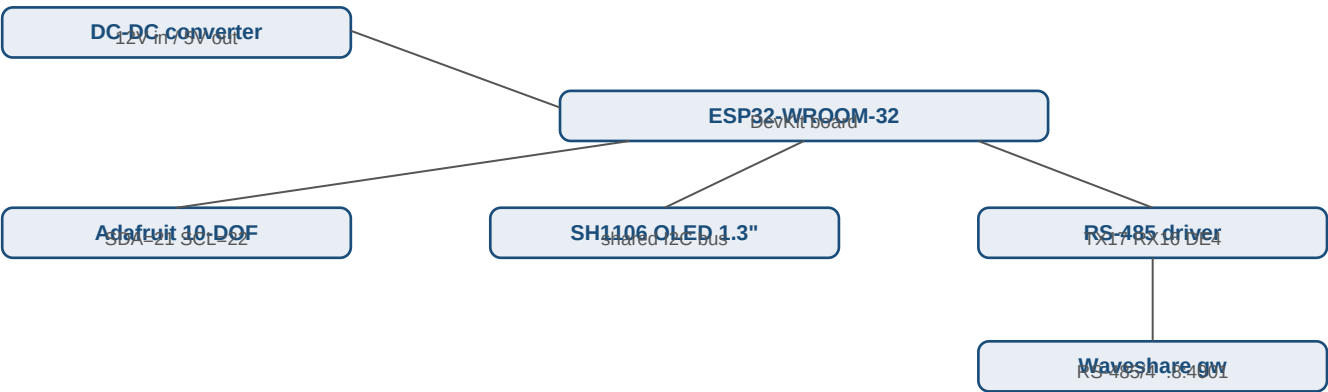


RVTC IMU Leveling Node — Build Sheet (HW-25)

Print and check off each connection as you solder/wire it. Space for notes on the right.



Wiring checklist

	Group	From	To	Notes
☐	Power	DC-DC 5V OUT	ESP32 5V / VIN	
☐	Power	DC-DC GND	ESP32 GND	
☐	Power	ESP32 3V3	Adafruit 10-DOF VIN	board accepts 3.3-5V, either is fine
☐	Power	ESP32 GND	Adafruit 10-DOF GND	
☐	Power	ESP32 3V3	OLED VCC	
☐	Power	ESP32 GND	OLED GND	
☐	Power	ESP32 3V3/5V	RS-485 module VCC	CHECK module's logic voltage first, see caution below
☐	Power	ESP32 GND	RS-485 module GND	
☐	I2C	ESP32 GPIO21 (SDA)	Adafruit 10-DOF SDA	
☐	I2C	ESP32 GPIO22 (SCL)	Adafruit 10-DOF SCL	
☐	I2C	ESP32 GPIO21 (SDA)	OLED SDA (bus tap)	
☐	I2C	ESP32 GPIO22 (SCL)	OLED SCL (bus tap)	
☐	UART/RS-485	ESP32 GPIO17 (TX1)	RS-485 module DI	
☐	UART/RS-485	ESP32 GPIO16 (RX1)	RS-485 module RO	
☐	UART/RS-485	ESP32 GPIO4	RS-485 module DE + RE (tied)	
☐	RS-485 bus	RS-485 module A	Waveshare RS-485/4, terminal A	
☐	RS-485 bus	RS-485 module B	Waveshare RS-485/4, terminal B	

Caution — verify before powering on:

ESP32 GPIOs are 3.3V logic and NOT 5V tolerant. Many cheap TTL-RS485 modules run their TTL side at 5V by default. Confirm the module's DI/RO/DE/RE pins are 3.3V-compatible (or use a level shifter) before wiring them directly to GPIO16/17/4 — a 5V signal back into the ESP32 can damage it.

Notes / bench observations: